对称密码，对称何在?

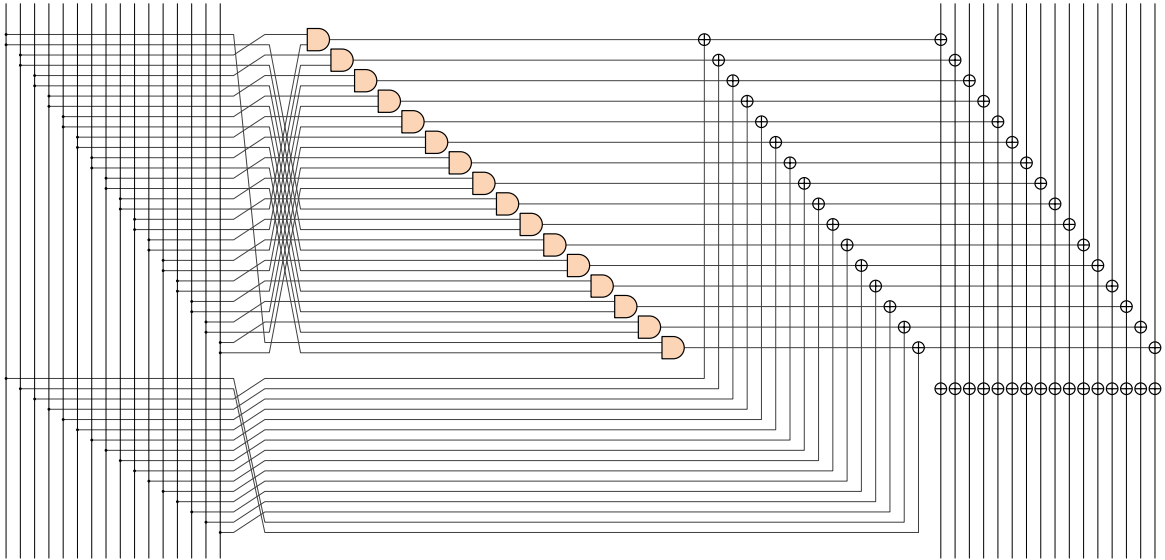
Tim Beyne

COSIC, KU Leuven

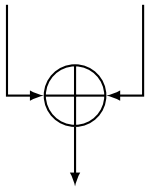
October 25, 2026



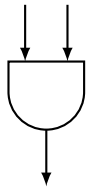
Symmetry in symmetric cryptography?



Symmetry in symmetric cryptography

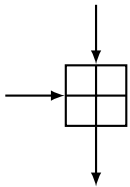


exclusive or

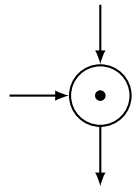


and

Symmetry in symmetric cryptography



addition



multiplication

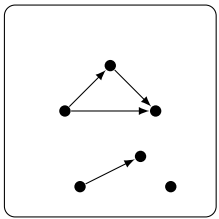
Monoid actions!

Overview

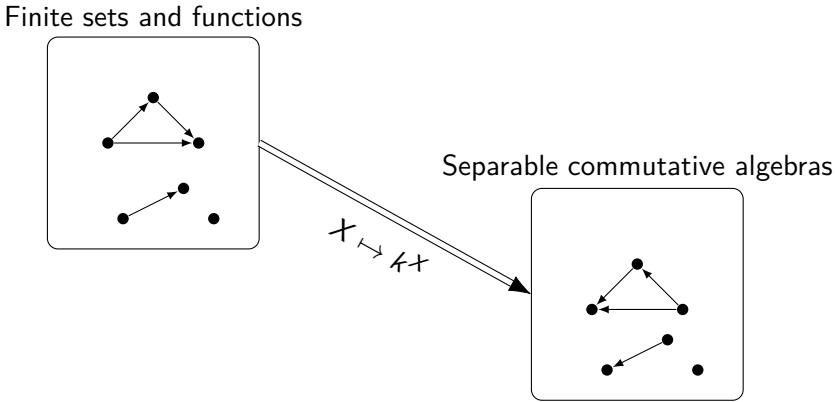
- Geometric approach to cryptanalysis
- Symmetry in cryptanalysis
- Security arguments

Geometric approach to cryptanalysis

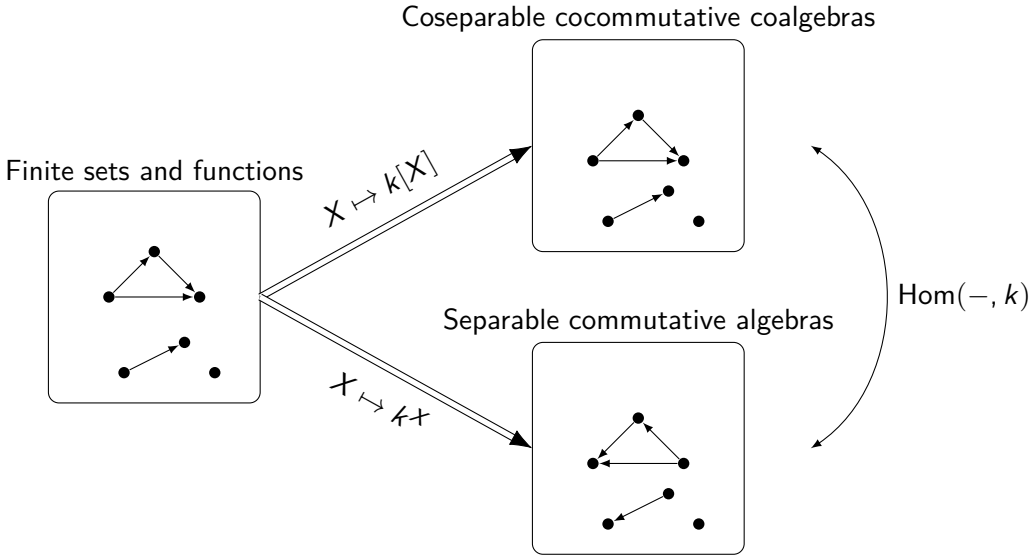
Finite sets and functions



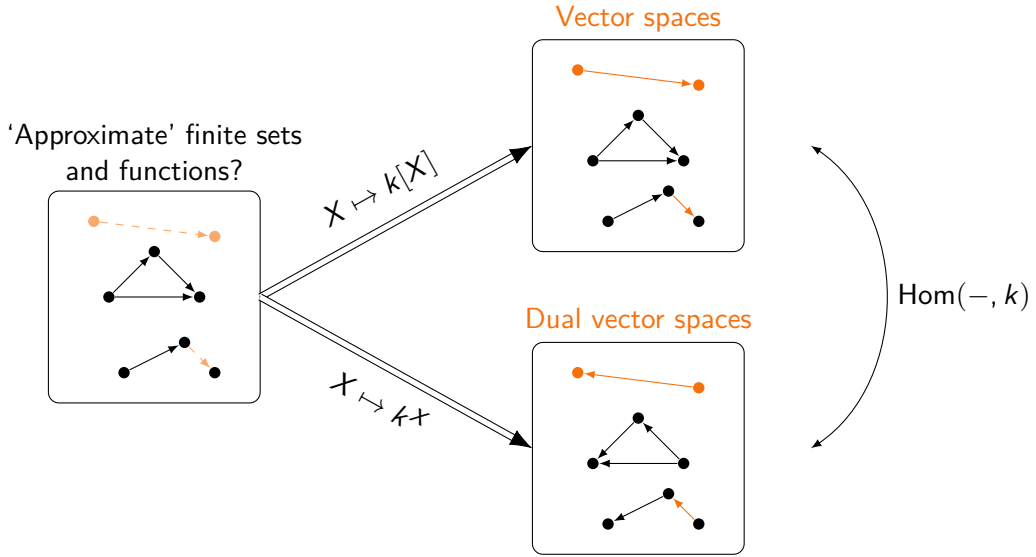
Geometric approach to cryptanalysis



Geometric approach to cryptanalysis

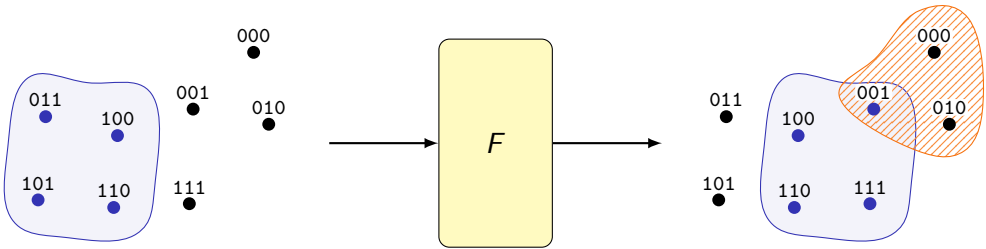


Geometric approach to cryptanalysis



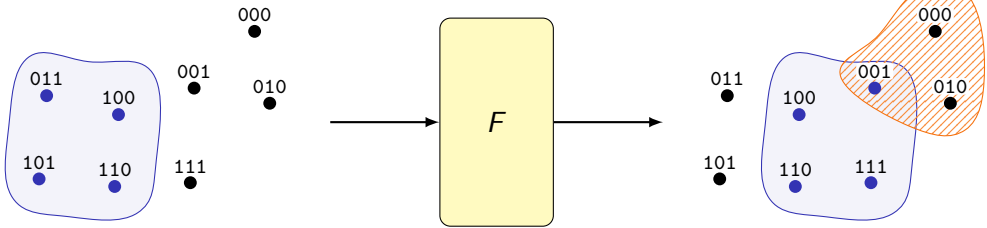
Cryptanalysis as linear algebra

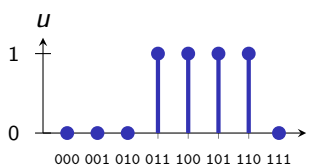
$F(x) = x + 010$ on $\mathbb{F}_2^3$

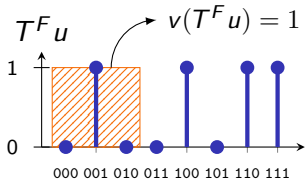


Cryptanalysis as linear algebra

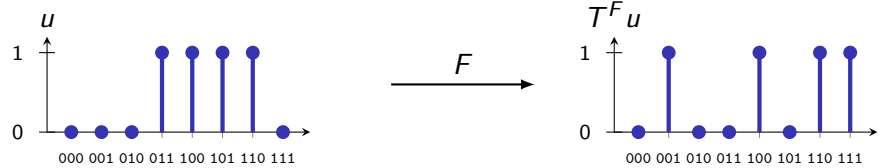
$F(x) = x + 010$ on $\mathbb{F}_2^3$

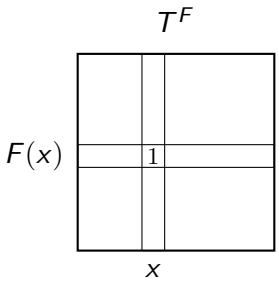




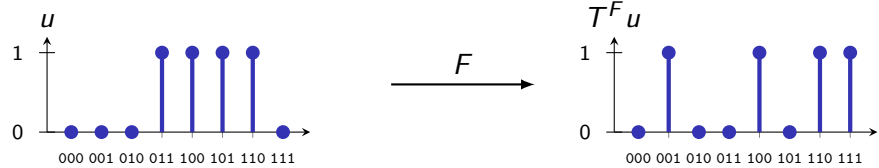


Propagation



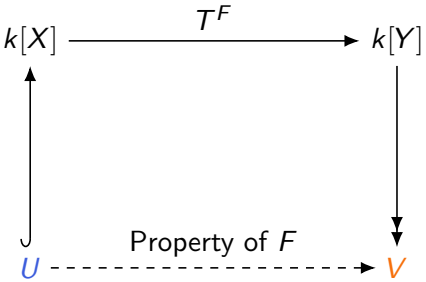


Propagation



$$T^F u = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

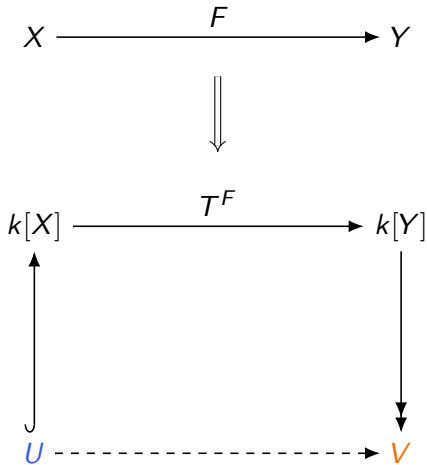
Cryptanalytic properties



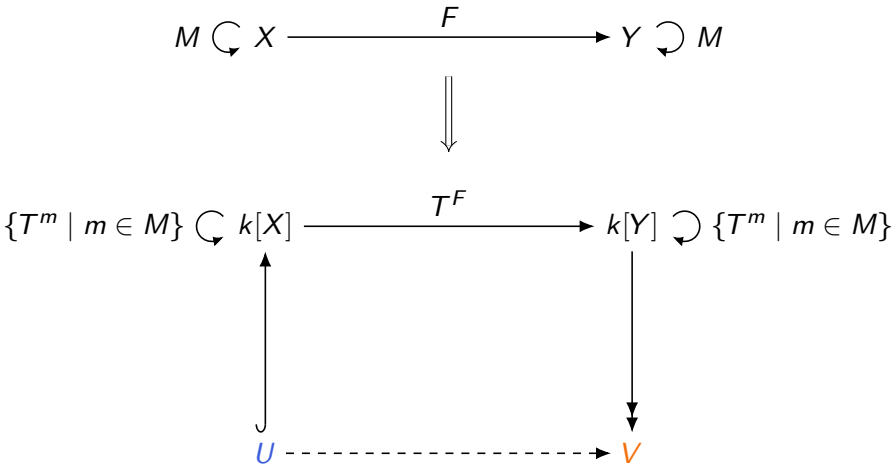
Overview

- Geometric approach to cryptanalysis
  Cryptanalysis as linear algebra
- Symmetry in cryptanalysis
- Security arguments

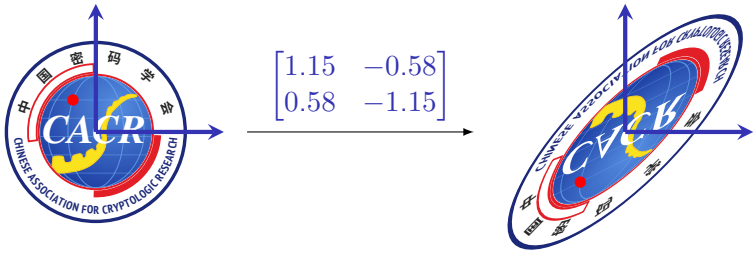
Monoid actions



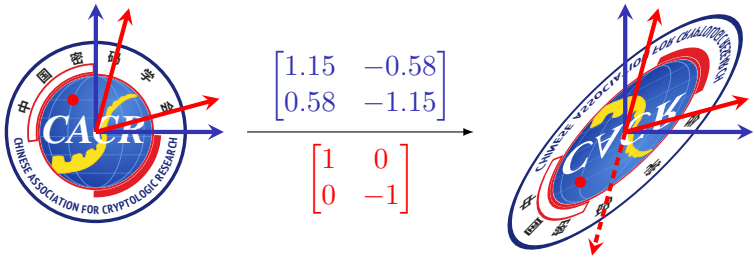
Monoid actions



Semisimple case



Semisimple case



Semisimple case

- M commutative and inverse

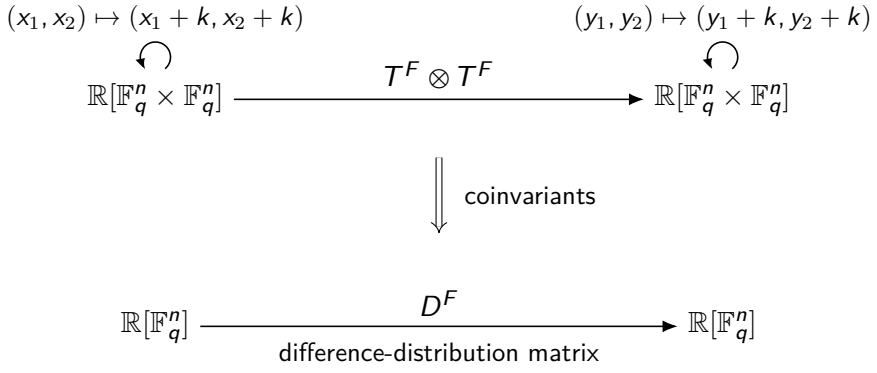
$$\widehat{M} = \mathrm{Hom}(M, k)$$

- Under suitable assumptions on k and on the action of M

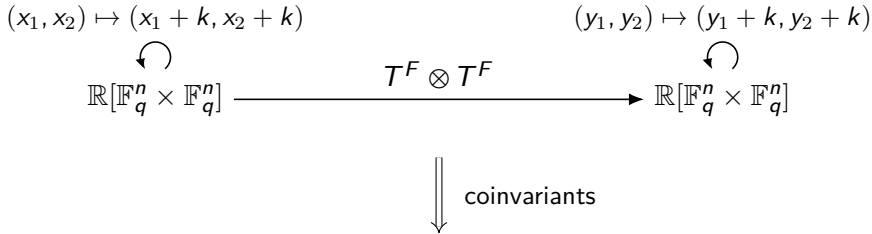
$$k[X] = \bigoplus_{\chi \in \widehat{M}} \mathrm{Span}\{b_\chi\} \text{ such that } T^m b_\chi = \chi(m)\, b_\chi$$

- Examples: linear cryptanalysis, (ultrametric) integral cryptanalysis

Pairwise properties



Pairwise properties



$$D^F_{b,a} = \Pr_{\mathbf{x}}\big[F(\mathbf{x} + a) = F(\mathbf{x}) + b\big]$$

Translation-invariant pairwise attacks

- Pairwise attacks using only invariant properties
  i.e. translation-invariant trails
- For a function $F = F_r \circ \cdots \circ F_2 \circ F_1$, fully characterized by

$$D^{F_r} \cdots D^{F_2} D^{F_1} \neq D^F$$

- Examples:
  - Differential cryptanalysis without quasidifferential trails
  - Linear cryptanalysis using squared correlations
  - ...

Overview

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Security arguments against translation-invariant pairwise attacks

- Difference-distribution matrix of a uniform random permutation:

$$U = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & \frac{1}{q^n-1} & \cdots & \frac{1}{q^n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \frac{1}{q^n-1} & \cdots & \frac{1}{q^n-1} \end{bmatrix}$$

- Security argument for $F = F_r \circ \cdots \circ F_2 \circ F_1$:

$$\left\| D^{F_r} \cdots D^{F_2} D^{F_1} - U \right\| \leq \varepsilon$$

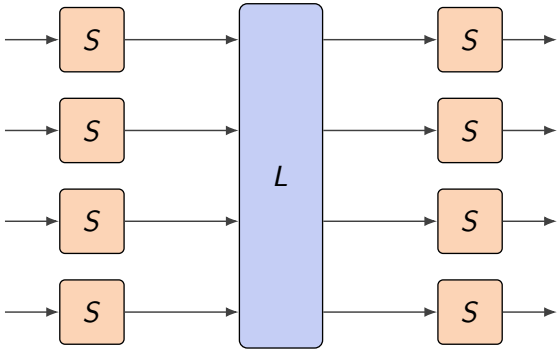
Information-theoretic interpretation

- $\|D^{E_k} - U\|_1 \leq \varepsilon$ implies $E_{\boldsymbol{k}}$ is $\varepsilon/2$-pairwise independent if $\boldsymbol{k}$ is uniform random
- Almost $\varepsilon/2$-pairwise independence of a block cipher $E_k \colon \mathbb{F}_q^n \to \mathbb{F}_q^n$
- Definition: for all (x_1, x_2) with $x_1 \neq x_2$,

$$\sum_{y_1 \neq y_2} \left| \Pr_{\boldsymbol{k}}\left[(E_{\boldsymbol{k}}(x_1), E_{\boldsymbol{k}}(x_2)) = (y_1, y_2)\right] - \frac{1}{q^n} \times \frac{1}{q^n - 1} \right| \leq \varepsilon$$

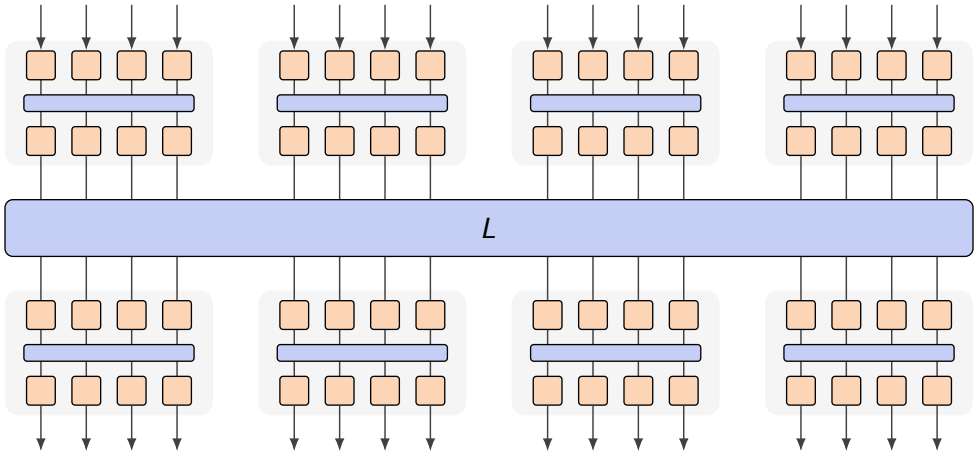
- Stronger security notion, but unrealistic assumptions

SHARK and four-round AES



$$\left(D^S\right)^{\otimes 4} D^L \left(D^S\right)^{\otimes 4}$$

SHARK and four-round AES



Proof strategy for SHARK

- Invariance of activity: $V = \mathrm{Span}\{\delta_0, \delta_{\mathbb{F}_q}\}$

$$D^S V \subseteq V$$

- Proof strategy: $\mathbb{R}[\mathbb{F}_q^n] = W_0 \oplus W_1 \oplus W_2 \oplus W_3 \oplus W_4$ with

$$W_l = \mathrm{Span} \bigcup_{\substack{I \subseteq \{1,\dots,n\} \\ |I| = l}} \left\{ v_1 \otimes v_2 \otimes v_3 \otimes v_4 \;\middle|\; \begin{array}{ll} v_i \in V^\perp & \text{if } i \in I \\ v_i \in V & \text{else} \end{array} \right\}$$

Proof strategy for SHARK

$$\|D-U\|_2 = \left\| \begin{bmatrix} \pi_{W_0}(D-U)\, i_{W_0} & \pi_{W_0} D\, i_{W_1} & \cdots & \pi_{W_0} D\, i_{W_n} \\ \pi_{W_1} D\, i_{W_0} & \pi_{W_1} D\, i_{W_1} & \cdots & \pi_{W_1} D\, i_{W_n} \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{W_n} D\, i_{W_0} & \pi_{W_n} D\, i_{W_1} & \cdots & \pi_{W_n} D\, i_{W_n} \end{bmatrix} \right\|_2$$

Proof strategy for SHARK

$q = 2^{32}, \quad \sigma = \|D^S - U\|_2$

$$\|D - U\|_2 \leq \left\| \begin{bmatrix} 2^{-61.4} & 0 & 2^{-28.0}\sigma^2 & 2^{-12.0}\sigma^3 & \sigma^4 \\ 0 & 2^{-27.3}\sigma^2 & 2^{-10.7}\sigma^3 & \sigma^4 & \sigma^5 \\ 2^{-28.0}\sigma^2 & 2^{-10.7}\sigma^3 & \sigma^4 & \sigma^5 & \sigma^6 \\ 2^{-12.0}\sigma^3 & \sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 \\ \sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 & \sigma^8 \end{bmatrix} \right\|_2 \leq 2^{-39}$$

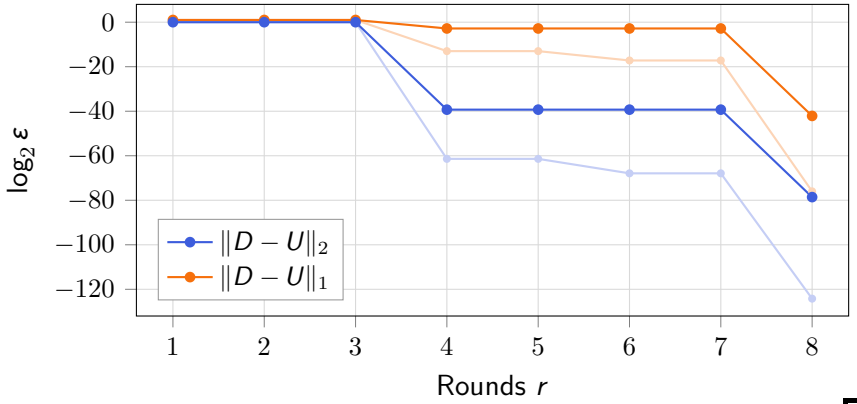
Proof strategy for SHARK

$q = 2^{32}, \quad \sigma = \|D^S - U\|_2, \quad \tau = \|D^S - U\|_F$

$$\|D - U\|_2 \leq \left\| \begin{bmatrix} 2^{-61.4} & 0 & 2^{-43.8}\sigma\tau & 2^{-44.5}\sigma\tau & 2^{-46.0}\sigma\tau \\ 0 & 2^{-43.6}\sigma^2 & 2^{-27.0}\sigma^3 & 2^{-12.0}\sigma^4 & \sigma^5 \\ 2^{-43.8}\sigma\tau & 2^{-27.0}\sigma^3 & 2^{-11.8}\sigma^4 & \sigma^5 & \sigma^6 \\ 2^{-44.5}\sigma\tau & 2^{-12.0}\sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 \\ 2^{-46.0}\sigma\tau & \sigma^5 & \sigma^6 & \sigma^7 & \sigma^8 \end{bmatrix} \right\|_2 \leq 2^{-61}$$

https://eprint.iacr.org/2026/2304

Collaborative research project: AES



- `https://cryptanalysis.info/aes-pairwise`
- LLM submissions ideally come with a formalization in Lean 4

Conclusion

- Geometric approach to cryptanalysis
  Cryptanalysis as linear algebra
- Symmetry in cryptanalysis
  Monoid actions
- Security arguments
  Translation-invariant pairwise attacks, $\|D - U\| \leq \varepsilon$